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9×19mm Parabellum

The **9×19mm Parabellum** (also known as the **9mm Parabellum** or **the 9mm Luger** or simply as the **9mm**) is a rimless, tapered firearms cartridge.

Originally designed by Austrian firearm designer Georg Luger in 1901,^[6] it is widely considered the most popular handgun and submachine gun cartridge due to its low cost, adequate stopping power and extensive availability.^{[7][8][9]}

Since the cartridge was designed for the Luger semi-automatic pistol, it has been given the designation of **9mm Luger** by the Sporting Arms and Ammunition Manufacturers' Institute (SAAMI)^[10] and the *Commission internationale permanente pour l'épreuve des armes à feu portatives* (CIP).^[2]

A 2007 US survey concluded that "about 60 percent of the firearms in use by police are 9mm [Parabellum]" and credited 9×19mm Parabellum pistol sales with making semiautomatic pistols more popular than revolvers.^[11]

Origins

The cartridge was developed by Austrian firearm designer Georg Luger in 1901. The cartridge was derived from an earlier round designed by Luger (7.65×21mm Parabellum), which itself was derived from a cartridge used in the Borchardt C-93 pistol (7.65×25mm Borchardt). Shortening the length of the cartridge case used in the Borchardt pistol allowed Luger to improve the design of the toggle lock and to incorporate a smaller, angled grip.

Luger's work on the Borchardt design evolved into the Luger pistol, which was first patented in 1898 and chambered in 7.65×21mm Parabellum. Demand from Germany for a larger caliber in their military sidearm led Luger to develop the 9×19mm Parabellum cartridge for the eventual P08 pistol. This was achieved by removing the bottleneck shape of the 7.65×21mm Parabellum case, resulting in a tapered rimless cartridge encasing a bullet that was 9 millimeters in diameter.

In 1902, Luger presented the new round to the British Small Arms Committee, as well as three prototype versions to the U.S. Army for testing at Springfield Arsenal in mid-1903. The Imperial German Navy adopted the cartridge in 1904, and in 1908, the German Army adopted it as well.^[7]

To conserve lead during World War II in Germany, the lead core was replaced by an iron core encased with lead. This bullet, identified by a black bullet jacket, was designated as the 08 mE (*mit Eisenkern*—"with an iron core"). By 1944, the black jacket of the 08 mE bullet was dropped, and these bullets were produced with normal copper-colored jackets. Another wartime variation was designated the 08 sE

9×19mm Parabellum	
	
Left to right; 9×19mm Parabellum, .40 S&W, 10mm Auto, .45 ACP	
Type	Pistol
Place of origin	German Empire
Service history	
Used by	Most military and civilian agencies around the world
Wars	World War I – present
Production history	
Designer	Georg Luger
Designed	1901
Produced	1902–present
Variants	9×19mm Parabellum +P
Specifications	
Parent case	7.65×21mm Parabellum
Case type	Rimless, tapered
Bullet diameter	9.01 mm (0.355 in)
Land diameter	8.82 mm (0.347 in)
Neck diameter	9.65 mm (0.380 in)
Base diameter	9.93 mm (0.391 in)
Rim diameter	9.96 mm (0.392 in)
Rim thickness	1.27 mm (0.050 in)
Case length	19.15 mm (0.754 in)
Overall length	29.69 mm (1.169 in)
Case capacity	0.862 cm ³ (13.30 gr H ₂ O)
Rifling twist	250 mm (1-9.84in)
Primer type	Berdan or Boxer small pistol

bullet and identified by its dark gray jacket, and was created by compressing iron powder at high temperature into a solid material (*Sintereisen*—"sintered iron").^[12]

The name *Parabellum* is derived from the Latin motto of *Deutsche Waffen- und Munitionsfabriken* (DWM), *Si vis pacem, para bellum* ("If you want peace, prepare for war").^{[13][14]}

Popularity

After the end of *World War I*, the popularity of the 9×19mm Parabellum cartridge increased with the vast number of *pistols* and *submachine guns* being adopted by many militaries and law enforcement agencies around the world.^[15] The 9×19mm Parabellum has become the most common caliber used by U.S. law enforcement agencies, primarily due to the availability of controllable compact pistols with large magazine capacities that use the cartridge.^[16]

From the 1980s to the 1990s, a sharp increase in popularity occurred with *semi-automatic pistols* in the United States by both law enforcement and military personnel, a trend foreshadowed by the adoption of the *Smith & Wesson Model 39* by the *Illinois State Police* in 1968. In addition, the *Beretta M9* (a military version of the *Beretta Model 92*) was adopted by the U.S. Army in 1985. Previously, most American police departments issued *.38 Special* and *.357 Magnum* caliber revolvers with a five-round or six-round capacity. The *.38 Special* was preferred to other weapons, such as variants of the *M1911*, because it offered low recoil, was small and light enough to accommodate different shooters, and was inexpensive to purchase.^[17] The 9×19mm cartridge is ballistically superior to the *.38 Special* revolver cartridge,^[18] is shorter overall, and being an autoloader cartridge, it is stored in flat magazines, as opposed to cylindrical speedloaders. This, coupled with the advent of the so-called "*wonder nines*", led to many U.S. police departments exchanging their revolvers for some form of 9mm semiautomatic pistols by the late 20th century.^[17]

In 2013, a chart of popular calibers that was released by the website *Luckygunner.com* showed 9×19mm Parabellum as having 21.4% of the entire cartridge market, followed by the *.223 Remington* at 10.2% (with 5.56 mm included this is 15.7%). The next most popular caliber was *.45 ACP*.^[19]

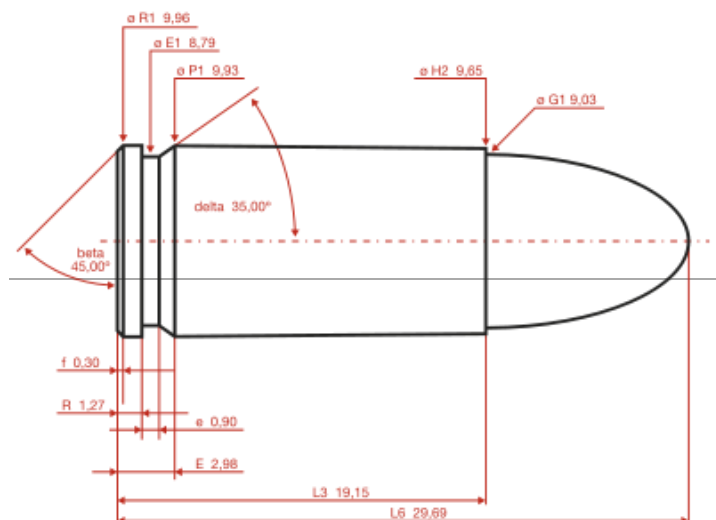
Cartridge dimensions

The 9×19mm Parabellum has 0.86 ml (13.3 grains H₂O) of cartridge case capacity.

Maximum pressure (CIP)	235.00 MPa (34,084 psi)	
Maximum pressure (SAAMI)	241.3165 MPa (35,000.00 psi)	
Ballistic performance		
Bullet mass/type	Velocity	Energy
7.45 g (115 gr) Federal FMJ	1,180 ft/s (360 m/s)	355 ft·lbf (481 J)
8.04 g (124 gr) Federal FMJ	1,150 ft/s (350 m/s)	364 ft·lbf (494 J)
8.04 g (124 gr) Norma ENVY FMJ	1,345 ft/s (410 m/s)	498 ft·lbf (675 J)
7.45 g (115 gr) Underwood JHP +P+	1,400 ft/s (430 m/s)	501 ft·lbf (679 J)
3.9 g (60 gr) RBCD TFSP	2,010 ft/s (610 m/s)	538 ft·lbf (729 J)
Test barrel length: 118mm (4.65")		
Source(s): Sellier & Bellot, ^[1] CIP, ^[2] Norma Ammunition, ^[3] Underwood Ammunition, ^[4] RBCD Performance Plus Ammunition ^[5]		



Three projectile types: unjacketed (lead), full metal jacket, and hollow point



9×19mm Parabellum maximum CIP cartridge dimensions:^[2] All sizes are given in millimeters (mm).

The cartridge headspaces on the mouth of the case.^[20] The common rifling twist rate for this cartridge is 250 mm (1 in 9.84 in), six grooves, ϕ lands = 8.82 mm, ϕ grooves = 9.02 mm, land width = 2.49 mm and the primer type is small pistol.

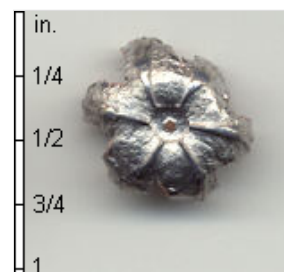
According to CIP rulings, the 9×19mm Parabellum cartridge case can handle up to 235.00 MPa (34,084 psi) $P_{\max}$ piezo pressure. In CIP-regulated countries, every pistol cartridge combination has to be proofed at 130% of this maximum CIP pressure to certify for sale to consumers. This means that 9×19mm Parabellum chambered arms in CIP-regulated countries are currently (2014) proof tested at 305.50 MPa (44,309 psi) PE piezo pressure.^[2]

The SAAMI pressure limit for the 9×19mm Parabellum is set at 241.32 MPa (35,001 psi) piezo pressure.^[21]

Performance

The round was originally designed to be lethal to 50 metres (160 ft), but is still lethal at longer ranges.^[22] The 9×19mm Parabellum cartridge combines a flat trajectory with moderate recoil. According to the 1986 book *Handloading*, "the modern science of wound ballistics has established beyond reasonable doubt that the 9mm cartridge is highly effective."^[23]

In 2014, the United States Federal Bureau of Investigation (FBI) released a report detailing the potential combat effectiveness of the 9×19mm Parabellum cartridge when compared to other calibers such as the .40 S&W and the .45 ACP cartridges that were specifically developed for use by the FBI.^[24] The report indicated that the new powders and more advanced bullet designs used in current 9×19mm Parabellum defensive loads allowed for the caliber to deliver adequate performance compared to other calibers, like the .40 S&W and .45 ACP. In addition to this, the lower recoil, less wear, cheaper ammunition, and higher capacity were all reasons that the report cited for the recent surge in orders of the ammunition from various police agencies. With a wider selection of officers able to shoot handguns chambered in 9×19mm Parabellum, many departments chose this caliber to standardize a single firearm and loading, making logistics and supply easier. Due to all these factors, law enforcement orders of 9×19mm Parabellum ammunition from all major ammunition manufacturers have risen significantly.^{[25][26]}



An expanded 124-grain 9×19 mm Parabellum jacketed hollow point

Improvements and variations

NATO standard

The 9mm is called NATO because it has become a standard pistol caliber for NATO and other military forces worldwide.^[27] The cartridge has been manufactured by, or for, more than 70 countries and has become a standard pistol caliber for NATO and other military forces around the world. Its official nomenclature among NATO members is "**9mm NATO**".^[9]

9mm NATO can be considered an overpressure variant of 9×19mm Parabellum that is defined by NATO standards. The proof pressure is 3,150 bars (45,700 psi), corresponding to a maximum service pressure of 2,520 bars (36,500 psi) (both using CIP methodology).^[28]

While the NATO standards do not specify the type of bullet to be used, Declaration III of the Hague Convention of 1899 prohibits the use of expanding ammunition in warfare by signatories, so official NATO 9mm ammunition is FMJ "ball" bullets.^[29] Declaration III does not apply in conflicts involving non-signatories to the Hague Convention, including paramilitary and other nongovernmental fighting forces.^[30]

In 2021, with the standardization of STANAG-4509, NATO adopted the 5.7x28mm SMG/pistol cartridge for the PDW program.^[31] It is in the same power range as 9x19mm Parabellum, but has a few additional advantages such as a larger magazine capacity and improved armor penetration.

Swedish m/39

9mm Parabellum entered Swedish service as m/39 with the import of the *Kulsprutepistol* m/39 from Austria, with a bullet weight of 7.5 grams (116 gr).^[32] During the Congo Crisis, the Swedish UN-contingent issued complaints about the performance of the m/39 cartridge (regular 9mm Parabellum) used. This resulted in a commission of the Swedish Army establishing in 1962 that a new round was needed for the *Carl Gustav* m/45. The resulting m/39B had a tombac-plated steel jacket surrounding the lead core. While the lands of the barrel can cut into the tombac, the steel jacket resists deformation, thus causes the gas pressure to rise higher than the previous soft-jacketed m/39, giving the 7.0 grams (108 gr) bullet a V_0 of 420 m/s (1,378 ft/s)^[33] and an impact energy of 600 joules. The mantle also acts like a penetrator when striking a target, going through up to 50 layers of kevlar, 7 cm of bricks, or 25 cm of wood, allowing the bullet to defeat body armor up to Type IIIA.

+P variant

Attempts to improve the cartridge's ballistics came in the early 1990s with the widespread availability of high-pressure loadings of the 9mm cartridge. Such overpressure cartridges are labeled "+P" (38,500 psi) or in the case of very high pressure loadings, "+P+" (42,000 psi).^[34] Velocity of these rounds is improved over standard loadings. In addition, improvements in jacketed hollow-point bullet technology have produced bullet designs that are more likely to expand and less likely to fragment than earlier iterations, giving a 9mm bullet better terminal effectiveness.^[35]

SESAMS

SESAMS weapons or components are normally painted blue or marked to denote their inert status and avoid a potentially catastrophic mixup with live-fire weapons.^[36] This allows the armed forces to train with nearly identical equipment as used in real-life situations.^[37]

Russian military overpressure variants



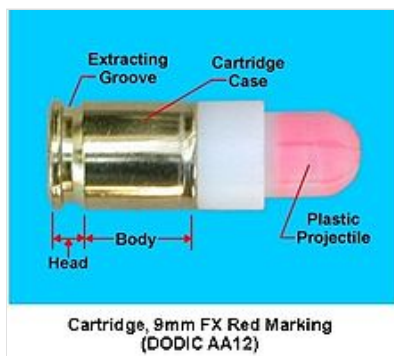
9mm live ammunition m/39 (left, with black seal) and m/39B (right, with red seal and a slightly more pointed shape)



9mm live ammunition m/39 and m/39B in their boxes



A 9mm Luger jacketed flat point cartridge variant



A red marking Simunition round

The Russian military has developed specialized 9×19mm cartridges that use relatively light bullets at high muzzle velocities for both pistols and submachine guns to defeat body armor.^[38]

Besides enhanced penetration capabilities, these overpressure variants offer a flatter trajectory and lessened recoil. The increase in service pressure causes a rise in bolt thrust, so this overpressure ammunition induces more stress on critical weapon parts during firing. After initial research conducted in

the late 1980s under the codename "Grach", the Russian armed forces adopted two specialized 9×19mm variants.^{[39][40]}

	7H21 (7N21)	7H31 (7N31) / PBP
Cartridge weight	9.5 g (147 gr)	8.1 g (125 gr)
Bullet weight	5.2 g (80.2 gr)	4.1 g (63.3 gr)
Muzzle velocity	460 m/s (1,509 ft/s)	550 m/s (1,804 ft/s)
Muzzle energy	561 J (414 ft·lbf)	620 J (457 ft·lbf)
Accuracy of fire at 25 m (27 yd) (R ₅₀)	25 mm (1.0 in)	
Maximum pressure	280 MPa (41,000 psi)	

- R₅₀ at 25 m (27 yd) means the closest 50 percent of the shot group will all be within a circle of 25 mm (1.0 in) radius at 25 m (27 yd).

The 7N21 (Cyrillic: 7H21) 9×19mm overpressure variant features an armor-piercing bullet and generates a peak pressure of 280 MPa (41,000 psi).^[39] The 7N21 bullet features a hardened (sub-caliber) steel penetrator core, enclosed by a bimetal jacket. The space between the core and jacket is filled with polyethylene, and the tip of the penetrator is exposed at the front of the bullet to achieve better penetration. The penetration range for body armor is specified at up to 40 m (130 ft). The MP-443 Grach and GSh-18 pistols and PP-19 Vityaz, PP-90M1 and PP-2000 submachine guns were designed for use with this overpressure cartridge. *Jane's Infantry Weapons* stated in 2003 that the 7N21 cartridge combined the 9×19mm Parabellum dimensions with a 9×21mm Gyurza bullet design and was developed specifically for the penetration of body armor and for the MP-443 Grach pistol, the latest Russian service pistol.^[41]

The 7N31 (Cyrillic: 7H31) / PBP 9×19mm overpressure variant uses the same concept with a similar but lighter bullet that achieves higher muzzle velocity. The penetration of an 8 mm (0.31 in)-thick St3 steel plate is specified at up to 10 m (33 ft).^[42] The 7N31 cartridge was developed in the late 1990s for the GSh-18 pistol. The 7N31 was adopted for the PP-90M1 and PP-2000 submachine guns. Its maximum service pressure remains unclear.

The construction of the two rounds allows them to be effective against both unarmored and armored targets. If the bullet strikes an unarmored target, it holds together to produce a wide wound channel. If the bullet strikes an armored target, the sleeve is stripped away, and the core penetrates alone. The disadvantage of the rounds is that high impact velocities are needed to work effectively, so the bullets are relatively light to maximize their muzzle velocity. This means they lose velocity relatively quickly, limiting their effective range.^[43]

Other variants



A box of 9mm FX blue marking (DODIC AA21) cartridges with a modified Beretta M9 pistol

VBR-B produces specialized bullets for this cartridge, a two-part controlled fragmenting projectile and an armor-piercing bullet that features a brass sabot and a hardened steel penetrator. These are designed to increase the content of the permanent wound cavity and double the chance of hitting a vital organ.^[44]

U.S. data

The energy delivered by most 9mm loads allows for significant expansion and penetration with premium hollow-point bullets. Illinois State Police, Border Patrol, Federal Air Marshals, and United States Secret Service favored and used 115 gr (7.5 g) +P+ 9mm loads at 1,300 ft/s (400 m/s) for years with excellent results.^[35] Massad Ayoob has stated that the "Tried, Tested, and True" 115 gr (7.5 g) +P or +P+ is the best self-defense load in this caliber.^[35]

Manufacturer	Load	Bullet mass	Velocity	Energy	Expansion ^[45]	Penetration ^[45]	PC ^[45]	TSC ^[45]
Cor-Bon	JHP+P	7.5 g (115 gr)	410 m/s (1,350 ft/s)	630 J (465 ft·lb)	14 mm (0.55 in)	360 mm (14.2 in)	56 mL (3.4 cu in)	631 mL (38.5 cu in)
ATOMIC Ammo	JHP+P	8.0 g (124 gr)	400 m/s (1,300 ft/s)	630 J (465 ft·lb)	15 mm (0.60 in)	330 mm (13 in)	"N/A"	"N/A"
Speer	Gold Dot JHP	8.0 g (124 gr)	350 m/s (1,150 ft/s)	494 J (364 ft·lb)	18 mm (0.70 in)	337 mm (13.25 in)	84 mL (5.1 cu in)	616 mL (37.6 cu in) (est) ^[46]
Federal	HydraShok JHP +P+	8.0 g (124 gr)	360 m/s (1,170 ft/s)	511 J (377 ft·lb)	17 mm (0.67 in)	340 mm (13.4 in)	77 mL (4.7 cu in)	734 mL (44.8 cu in) ^[46]
Remington	Golden Saber JHP	9.5 g (147 gr)	300 m/s (990 ft/s)	430 J (320 ft·lb)	16 mm (0.62 in)	370 mm (14.5 in)	72 mL (4.4 cu in)	544 mL (33.2 cu in)
Winchester	Silvertip	7.5 g (115 gr)	373 m/s (1,225 ft/s)	519 J (383 ft·lb)	18 mm (0.72 in)	200 mm (8.0 in)	54 mL (3.3 cu in)	274 mL (16.7 cu in)
Winchester	WWB JHP	9.5 g (147 gr)	300 m/s (990 ft/s)	430 J (320 ft·lb)	15 mm (0.58 in)	400 mm (15.9 in)	69 mL (4.2 cu in)	321 mL (19.6 cu in)
Winchester	FMJ	7.5 g (115 gr)	352 m/s (1,155 ft/s)	462 J (341 ft·lb)	9.1 mm (0.36 in)	620 mm (24.5 in)	41 mL (2.5 cu in)	174 mL (10.6 cu in)

Key:

Expansion: expanded bullet diameter (ballistic gelatin)

Penetration: penetration depth (ballistic gelatin)

PC: permanent cavity volume (ballistic gelatin, FBI method)

TSC: temporary stretch cavity volume (ballistic gelatin)

See also

- 9 mm caliber
- 9mm Major
- List of firearms
- List of handgun cartridges
- List of rifle cartridges
- Table of handgun and rifle cartridges

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External links

- Article on 9×19 mm Parabellum cartridge collecting including history with photos and descriptions of variations including headstamps (<http://cartridgecollectors.org/documents/Introduction-to-9mm-Luger-Cartridges.pdf>)
 - Ballistics By The Inch 9×19 mm Parabellum Results. (<http://www.ballisticsbytheinch.com/9luger.html>)
 - Data on the Russian ammo (in Russian) (http://www.kalashnikov.ru/upload/medialibrary/6c0/24_30.pdf)
 - DIRECT FIRE AMMUNITION Handbook 2021, Project Manager Maneuver Ammunition Systems (https://jpeoaa.army.mil/Portals/94/MAS/Documents/MASHandbook_2021.pdf)
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